Flat anti weight

> The recipe of record sets the anti-subspace weight high on the early slices and low on the late ones. We tried the same weight at every slice instead. A weight of 0.17, the same at every slice and every epoch, met the bar we set in advance for doing about as well as the recipe, at eight seeds. A higher one cost removal° on some seeds. Lower ones each ended with one run in eight over the spill° criterion, which is too few to tell from the recipe.

Removal. The drop on `difference` under the full edit at every position and slice, as a share of the way from the clean score to the target null....

Spill. The largest drop on any other op, at the dose where it is largest....

The recipe of record anneals the anti-subspace weight from 0.25 on every slice to a hold that falls off with depth, from 0.2 at the embedding to 0.03 at slice 4. The falloff came from the *split-anti-seeds*° experiment, where both holds that were the same at every slice did worse. But that experiment used an older pull: a step on slices 2 to 4, with a sharp pooling temperature° from the start and no pool over slices.

split-anti-seeds. Under a step pull on slices 2 to 4, an anti hold that falls off with depth kept the anchor on all eight runs, while holds of 0.2 and 0.1 at every slice each lost one run to the query answer, and the high one removed less.

Pooling temperature. How sharply the anchor pull picks its positions....

Each flat hold there lost the anchor on one run, but the hold can't have caused those losses. The lost runs parted from the kept ones within the first sixty epochs (*loss-rate*°), while the anti weight was still near its start of 0.25 in every condition. The falloff also lost one run of forty this way, later on, and a soft start for the pooling temperature has since stopped such losses.

loss-rate. With a soft start for the pooling temperature, no run of forty lost the anchor, against one of forty without it.

Two other results stand against the flat holds: the high hold removed less on the seeds where the pull settled slowly, and the low one spilled. A sharp pool over slices 1 to 3 has since taken away the spill from block 1 (*soft-pool-scout*°), so a flat hold may now do as well, which would make the recipe simpler to describe.

soft-pool-scout. Keeping the pool over slices 1 to 3 at a sharp temperature removed the early spill, which entered through block 1.

This report trains the recipe at eight seeds under eight anti weights, with a control at each seed: 72 runs, in two rounds. The first round held 0.115 or 0.2 at every slice. The second added constant weights of 0.14 and 0.17 between them, and an anneal to 0.115 from a lower start of 0.17.

Two of the seven other weights meet the bar we set before the runs for doing about as well as the falloff (in Scope and method): `level 0.17`, a constant weight, and `from 0.17`. Both were chosen after the first round was read and are scored on the same seeds, and `level 0.17` meets the removal margin on its mean, with an interval that reaches past it. The two at 0.2 miss it on removal. `flat 0.115`, `level 0.115` and `level 0.14` miss it by one run over the spill criterion each, which is too few to tell them from the recipe.

Observations

- How the anchor settles (E1): every run of every condition is called kept at epoch 64, with or without the high start, and α° at the example answers settles about as fast in all of them.
- Removal° and spill at the end (E2): at 0.2 everywhere, no run spills° and the color embeddings stop leaning onto e_1, but two or three runs in eight remove much less, and one in each condition ends under the lost line. At 0.115 or 0.14 everywhere, removal stays near the recipe, and one run in each condition ends over the spill criterion. With no high start at 0.115, block 1 spills on more runs. `level 0.17` and `from 0.17` end with no run over and removal within a few hundredths of the recipe. Six runs miss the task gate, five of them on the mean over ops: each learned the HSV ops late and had flattened a little short of the controls by epoch 400.

Alignment. How far a state points along e_1...

Removal. The drop on `difference` under the full edit at every position and slice, as a share of the way from the clean score to the target null....

Spill. The largest drop on any other op, at the dose where it is largest....

- Along training (E3): at 0.2, three of the low runs remove about as much as the rest at epoch 64 and then slide, and two are low from the start and stay low, where the slow runs of the recipe catch up. Spill along training is rarest at 0.2 and most common at `level 0.115`, and the recipe crosses the spill criterion on more runs than any weight from 0.14 up.
- Where the anchor sits (E4): the op sits on e_1 mostly at slices 2 to 4, and the other ops lean on it most at slice 1, the same under every condition. The falloff is high where the other ops lean and low where the op sits.

Scope and method

This is an exploratory study, with no preregistration and no gate of its own. Every run is the recipe of record, `τ_ℓ fixed` as the *fixed-confirm*° experiment trained it, with one change: the anti weight.

fixed-confirm. `τ_ℓ fixed` kept the anchor on all eight runs at fresh seeds with removal near 0.9 and no run over the spill criterion, and became the recipe of record.

Conditions

`recipe` is `τ_ℓ fixed` itself, with the falloff over depth, retrained here so that every condition shares the card and the arithmetic. `flat 0.115` anneals from 0.25 to a hold of 0.115 at every slice. That is the mean of the falloff over the five slices, so the term has the same total weight, spread evenly. `flat 0.2` anneals to 0.2 at every slice. That is the falloff value at the embedding, which kept the color embeddings from standing in for the op (embedding-lean). `level 0.115` and `level 0.2` hold their value at every slice from the first epoch, with no high start; `level 0.2` is close to `flat 0.2`, which only starts at 0.25. A second round, after the first was read, added `level 0.14` and `level 0.17` between the two level weights, and `from 0.17`, which anneals to a hold of 0.115 at every slice from a start of 0.17 in place of 0.25.

Every condition keeps the end anneal it shares with the anchor, over the last 40 epochs. The anneal from 0.25 is slow to start, so the high start lasts: at epoch 64 the weight is still within a hundredth of 0.25 on every slice of every annealed condition, and at epoch 200 it is a little over halfway to the hold.

The two flat holds match the falloff on its mean and on its highest value. `flat 0.115` is lower than the falloff at slice 1, the same at slice 2, and higher at slices 3 and 4, and `flat 0.2` is higher at every slice but the embedding. No flat hold sits below the falloff on the slices the pull acts on, since 0.03 at every slice let the color embeddings lean onto e_1 in the *late-pull*° experiment.

late-pull. With the pull moved off the embedding, an anti hold of 0.03 at every slice let the color embeddings lean onto e_1 and block 1 spill, and higher holds reduced both at some cost to removal.

`level 0.115` and `from 0.17` ask a separate question, and their comparison is with `flat 0.115`, which differs only in the start. In the *ex-2.1.11*° experiment, an anti weight held at its low value from the start let the pull latch onto a syntax token. That low value was 0.3 times the anchor weight, where 0.115 is a little more than the anchor weight (0.1), so `level 0.115` is a milder case than that one. The soft start of the pooling temperature° now spreads the early pull over positions, and may do that job instead.

ex-2.1.11. In D2.1, an anti weight held at its low value from the start let the pull latch onto a syntax token, and one held at its high value throughout left the response less graded.

Pooling temperature. How sharply the anchor pull picks its positions....

The recipe is retrained at seeds 720 to 727, where no run of it had been trained past epoch 64 (the *loss-rate*° counts stopped there). We compare conditions as two samples of eight runs, with a 95% Welch t interval on the difference of seed means,[1] and Fisher's exact test for counts of runs (lost, or over the spill° criterion).

loss-rate. With a soft start for the pooling temperature, no run of forty lost the anchor, against one of forty without it.

1 Welch t interval...

Spill. The largest drop on any other op, at the dose where it is largest....

At eight runs a side, removal° gaps under about 0.05 are hard to see. But the seed range of the recipe (about 0.08 in *fixed-confirm*) suggests an interval of about ±0.03 on removal, so the experiment can bound what a flat hold costs.

Removal. The drop on `difference` under the full edit at every position and slice, as a share of the way from the clean score to the target null....

We take a flat hold to do about as well as the falloff when all of these hold: its removal is within 0.05 of the recipe, it loses no anchor, and no run is over the spill criterion.

Every run trained and was scored on L40S cards. The earlier runs of the recipe (704 to 719) trained on L4 cards with the determinism flags on, so they are left out.

The measurements

Every edit below is the e_1 edit. Removal° and spill° are each a drop under that edit, less a baseline: the median drop when the same edit is made along each of the other 63 coordinate axes of the run (the rule of the D2.2 design).

Removal. The drop on `difference` under the full edit at every position and slice, as a share of the way from the clean score to the target null....

Spill. The largest drop on any other op, at the dose where it is largest....

Removal

The drop on `difference` under the full edit at every position and slice, as a share of the way from the clean score to the target null. Higher is better, and a run below 0.5 has lost the anchor.

Spill

The largest drop on any other op, at the dose where it is largest. Lower is better, and a run above 0.02 is over the criterion. Along training (E3) it is at the full dose only.

Block-1 spill

Spill under the full edit at slice 1 alone, where the spill of earlier recipes entered.

Task change

The clean score per op, less the mean of the controls. The gate of the D2.2 design asks for a mean change over ops of at least −0.02, and at least −0.05 on any one op.

Call

The early verdict on whether a run kept the anchor, from mean α at the example answers of the `difference` probe contexts at slice 3, over epochs 44 to 64. A run below 0.15 is called lost, and one at 0.28 or above is called kept. The cuts were set on runs whose anti weight was near 0.25 over the window, so for the level conditions the call is descriptive and removal at the end decides.

Embedding lean

The correlation of the e_1 component of the color embeddings with their lightness. Near zero, no color embedding stands in for the op.

How the anchor settles (E1)

Does every condition settle on the example answers, and as quickly?

Up to the call° at epoch 64 the three conditions that start at 0.25 are the same recipe, so any losses there test the others alone. A level run that aligns a syntax token or the query answer would show up here.

Call. The early verdict on whether a run kept the anchor, from mean α at the example answers of the `difference` probe contexts at slice 3, over epochs 44 to 64....

A run of any condition can still lose the anchor after epoch 64. In the *loss-rate*° experiment, the slow runs of the older recipe stalled until the anti weight at slice 3 had fallen to about half. A hold of 0.2 never falls that far, so the speed of settling may separate `flat 0.2` from the rest.

***loss-rate*.** With a soft start for the pooling temperature, no run of forty lost the anchor, against one of forty without it.

The table gives the call, how quickly α° settled, and removal° at epoch 64 for each condition; the last two columns count runs from E3.

Alignment. How far a state points along e_1...

Removal. The drop on `difference` under the full edit at every position and slice, as a share of the way from the clean score to the target null....

condition	called kept ↑	α over epochs 44 to 64 ↑	epoch α reaches 0.4 ↓	removal at epoch 64 ↑	slid ↓	over along training ↓
`recipe`	8 of 8	0.56 (0.53 to 0.59)	25 (24 to 28)	0.70 (0.45 to 0.83)	0	3
`flat 0.115`	8 of 8	0.56 (0.50 to 0.58)	24 (24 to 24)	0.70 (0.36 to 0.88)	0	1
`from 0.17`	8 of 8	0.59 (0.57 to 0.63)	24 (24 to 24)	0.81 (0.43 to 0.88)	0	1
`flat 0.2`	8 of 8	0.57 (0.52 to 0.60)	26 (24 to 28)	0.79 (0.47 to 0.90)	2	0
`level 0.115`	8 of 8	0.63 (0.59 to 0.66)	24 (24 to 28)	0.84 (0.74 to 0.90)	0	4
`level 0.14`	8 of 8	0.61 (0.58 to 0.63)	24 (20 to 24)	0.86 (0.75 to 0.91)	0	2
`level 0.17`	8 of 8	0.58 (0.53 to 0.60)	24 (24 to 28)	0.77 (0.37 to 0.87)	0	1
`level 0.2`	8 of 8	0.58 (0.56 to 0.61)	25 (24 to 28)	0.82 (0.58 to 0.89)	1	1

How the anchor settles, and spill along training. *The call and its window mean of α at the example answers at slice 3; the first record at which that α reaches 0.4; removal at epoch 64 (seed mean and range); the runs whose removal at the end is more than 0.1 under their best; and the runs whose spill is over the criterion at any kept checkpoint or at the end.*

All 64 runs are called kept, by a wide margin, and α at the example answers reaches 0.4 by epoch 28 on every run. The level conditions settle as quickly as the rest, so the early pull finds the example answers without the high start; nothing latched onto a syntax token or the query answer. The window mean of α is higher the lower the weight over the window, from 0.63 at `level 0.115` to 0.56 on the recipe, which fits a weaker push from e_1.

The time to settle does not separate the conditions either, so whatever sets `flat 0.2` apart comes later (E3). With every run called kept, the low-removal runs of E2 are runs whose α settled on time, and whose removal fell behind or never caught up. Two of them end under 0.5 and so count as lost at the end; the call missed them.

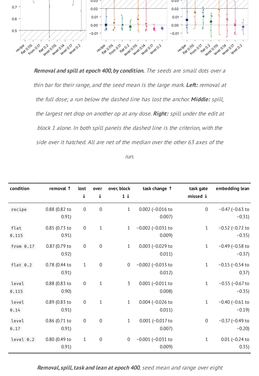
Removal and spill at the end (E2)

Does a flat hold remove as much, and spill as little? Block-1 spill and the embedding lean say where any spill comes from.

Spill. The largest drop on any other op, at the dose where it is largest.

Block 1 spill. Spill under the flat hold, on its own axes, where the spill of 0.2 begins to fall.

Embedding lean. The combination of the six components of the color embeddings with their lightness.



Removal and spill at epoch 400, by condition. *The seeds are small dots over a thin bar for their range, and the seed mean is the large mark. **Left:** removal: at the full dose; a run below the dashed line has lost the anchor. **Middle:** spill, the largest net drop on another op at any dose. **Right:** spill under the edit at block 1 alone. In both spill panels the dashed line is the criterion, with the side over it hatched. All are net of the median over the other 63 axes of the run.*

condition	removal ↑	lost	over	over, block 1	task change ↑	task gate missed ↓	embedding lean
recipe	0.88 (0.82 to 0.91)	0	0	1	0.002 (-0.016 to 0.007)	0	-0.47 (-0.63 to -0.31)
flat 0.115	0.85 (0.73 to 0.91)	0	1	1	-0.002 (-0.011 to 0.009)	1	-0.52 (-0.72 to -0.35)
from 0.17	0.87 (0.79 to 0.92)	0	0	1	0.003 (-0.029 to 0.011)	1	-0.49 (-0.58 to -0.37)
flat 0.2	0.78 (0.44 to 0.91)	1	0	0	-0.002 (-0.013 to 0.012)	1	-0.55 (-0.54 to 0.37)
level 0.115	0.88 (0.83 to 0.90)	0	1	5	0.001 (-0.011 to 0.008)	1	-0.55 (-0.67 to -0.35)
level 0.14	0.89 (0.83 to 0.91)	0	1	1	0.004 (-0.026 to 0.011)	1	-0.40 (-0.61 to -0.19)
level 0.17	0.86 (0.71 to 0.91)	0	0	1	0.001 (-0.017 to 0.007)	0	-0.37 (-0.49 to -0.20)
level 0.2	0.80 (0.49 to 0.91)	1	0	0	-0.001 (-0.011 to 0.009)	1	0.01 (-0.24 to 0.15)

Removal, spill, task and lean at epoch 400. *seed mean and range over eight runs, or a count of runs of eight. "Lost" counts runs with removal under 0.5. "Over" counts runs over the spill criterion under the full edit, and "over, block 1" under the edit at block 1 alone. The task change is the mean over ops of the clean score less the mean of the eight controls. The embedding lean is the correlation of the ev component of the color embeddings with their lightness.*

A weight of 0.2 everywhere is the cleanest: no run over the criterion, under the full edit or at block 1, and the color embeddings barely lean onto ev. But removal says.

Removal. The drop on *birthyear*, under the full edit, at every step: the seeds, the seed mean, and the range. The dashed line is the criterion: a run below it has lost the anchor.

Three runs of `flat 0.2` and two of `level 0.2` end well below the lowest run of the recipe, and the lowest in each (0.44 and 0.49) is under the line where we call the anchor lost, the second only just. That is the same pattern the high hold showed under the older pull, so it does not seem to depend on how the pull is pooled.

Cut. The early verdict on whether a ramp let the anchor. From *travel* at all 8 checkpoints, the mean of 64 reference probe contexts at block 1, over epochs 96 to 256.

A weight of 0.115 everywhere keeps removal close to the recipe. `level 0.115` matches it, and `flat 0.115` is a little lower: four of its runs end under the lowest of the recipe, two of them by less than 0.01, and the lowest at 0.73.

Each condition at 0.115 ends with one run over the spill criterion, on an HSV op. At `flat 0.115` that run is seed 720, also the run of the recipe nearest the criterion (at 0.019), so on spill at the end the two conditions differ by about 0.01 on one seed. The embedding lean stays about where the recipe has it, and with no high start block 1 spills on more runs.

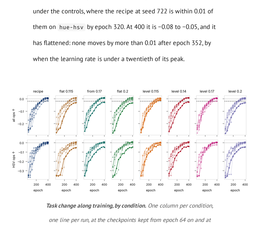
Between 0.115 and 0.2, `level 0.17` ends with no run over the spill criterion, its highest at 0.001, and removal at 0.86, 0.02 under the recipe on its mean, which is inside the margin, though the interval reaches 0.08 under. One run sits lower, at 0.71: seed 720, the lowest run of the recipe too, and the only run of `level 0.17` over at block 1 (0.029). `level 0.14` keeps removal at the recipe (0.89) and ends with one run over, on `mix`. from 0.17 ends with no run over and removal at 0.87. At block 1 it has 1 run over, just (0.022), against 5 at `level 0.115`, which would fit a start of 0.17 being enough to keep block 1 about as clean as the start of 0.25 does, though one run against three of eight is within what chance gives (Fisher's exact test, $p = 0.6$).

The recipe has removal as high as any condition, and no run over the criterion under the full edit at the end, though its run at seed 720 ends just under it. Its embedding lean is about as strong as at 0.115, though it holds 0.2 at the embedding, so the lean seems to follow the weight past the embedding more than the weight on it. Under the sharp pool, the lean costs the recipe no selectivity at the end.

condition	against	removal gap	95% interval, low	95% interval, high
flat 0.115	recipe	-0.055	-0.090	+0.021
from 0.17	recipe	-0.008	-0.044	+0.029
flat 0.2	recipe	-0.102	-0.215	+0.031
level 0.115	recipe	+0.001	-0.027	+0.029
level 0.14	recipe	+0.005	-0.024	+0.033
level 0.17	recipe	-0.024	-0.080	+0.032
level 0.2	recipe	-0.078	-0.204	+0.047
level 0.115	flat 0.115	+0.056	-0.019	+0.091
from 0.17	flat 0.115	+0.027	-0.031	+0.085
level 0.2	flat 0.2	+0.024	-0.142	+0.189

Removal against the recipe, and each condition with a lower start against the flat hold of the same value: *the difference of seed means and its 95% Welch t interval, eight runs a side.*

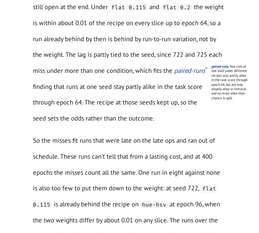
On removal the intervals are wide, since the lean runs at 0.2 spread the seed range. `level 0.115` sits within a few hundredths of the recipe either way. The counts of runs over the criterion (one against none) are too small for Fisher's exact test to tell apart ($p = 1.0$), and the gaps in mean spill have intervals that span zero, so the runs at the end do not separate the low weights from the recipe on spill.



Task change by op at epoch 400, by condition. *The clean score of each op less the mean of the eight controls, one small dot per run over a thin bar for the range, with the seed mean as the large mark. The dashed line is the gate on any one op (-0.05), with the side under it hatched; the gate on the mean over ops (-0.02) is not drawn.*

Six runs miss the task gate, one in each condition but the recipe and `level 0.17`. The worst op of each is an HSV op, but 5 of the six miss on the mean over ops (about -0.03, where the gate is -0.02), with every op a little under the controls, the ops learned early included. They share seeds: seed 722 under each condition that ends at a hold of 0.115 everywhere, seed 725 under both at 0.2, and seed 720 under `level 0.14`.

The HSV ops are the last the model learns, mostly between epochs 100 and 350, and these runs learned theirs later than the recipe did at the same seed. At epoch 256 the worst op of each is -0.14 to -0.11 under the controls, where the recipe at seed 722 is within 0.01 of them on `hue-hsv` by epoch 320. At most 40 is -0.08 to -0.05, and it has flattened: none moves by more than 0.01 after epoch 352, by when the learning rate is under a twentieth of its peak.



Task change along training, by condition. *One column per condition, one line per run, at the checkpoints kept from epoch 64 on and at the end; the runs that miss the gate at the end are drawn solid and the rest faint. **Top:** the clean score less the mean of the controls, as a mean over all seven ops; the dashed line is the gate on that mean (-0.02). **Bottom:** the same over the three HSV ops alone.*

The misses are runs that learned the whole task late. By epoch 64 each is the furthest behind of its condition on the mean over ops, or second furthest, and it stays behind on every op while the others close on the controls. The HSV ops, learned last, are where the gap is still open at the end. Under `flat 0.115` and `flat 0.2` the weight is within about 0.01 of the recipe on every slice up to epoch 64, so a run already behind by then is behind by run-to-run variation, not by the weight. The lag is partly tied to the seed, since 722 and 725 each miss under more than one condition, which fits the *paired runs* finding that runs at one seed stay partly alike in the task score through epoch 64. The recipe at those seeds kept up, so the seed sets the odds rather than the outcome.

paired runs. Two runs of one seed under different conditions share their start, so a run through epoch 64 but one only through epoch 32 is behind, and no more alike than chance in spill.

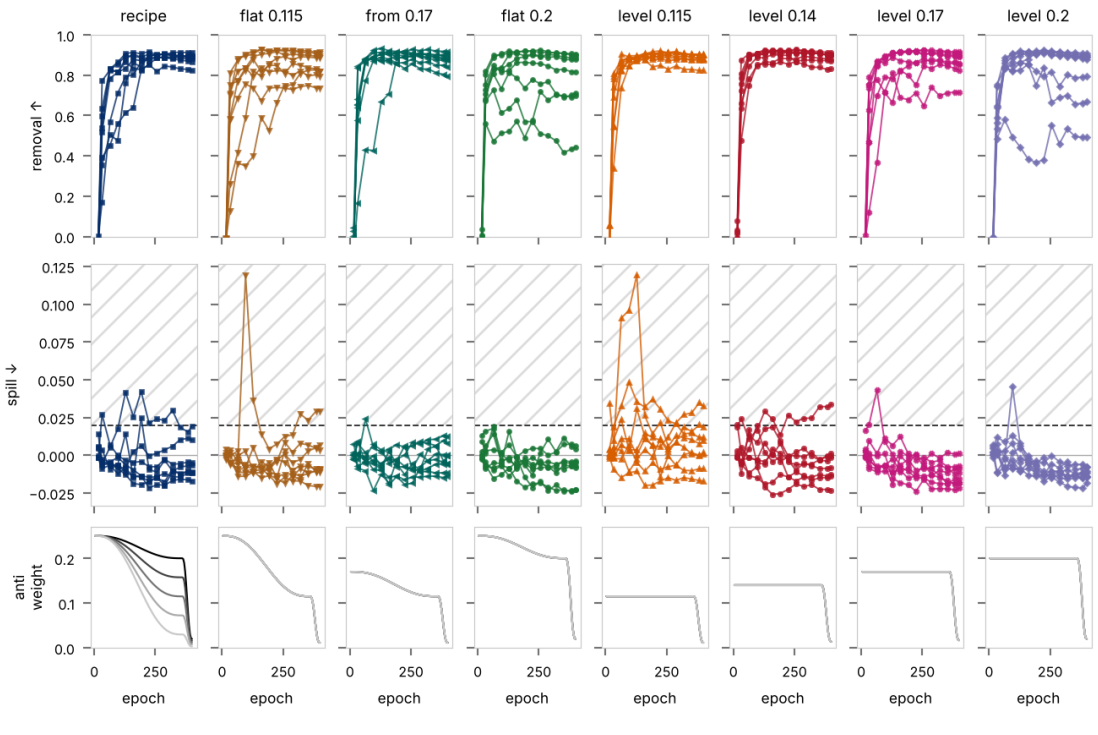
So the misses fit runs that were late on the late ops and ran out of schedule. These runs can't tell that from a lasting cost, and at 400 epochs the misses count all the same. One run in eight against none is also too few to put them down to the weight: at seed 722, `flat 0.115` is already behind the recipe on `hue-hsv` at epoch 96, when the two weights differ by about 0.01 on any slice. The runs over the spill criterion at 0.115 are on HSV ops too, which may be the same lateness, though that is two runs, and the run over at `level 0.14` is on `mix`.

Along training (E3)

The late spill° of earlier recipes rose as the anti weight annealed (*late-spill-scout*°), and a flat hold changes what the weight anneals to. Does spill rise late under any condition, and when do the low runs at 0.2 fall behind?

Spill. The largest drop on any other op, at the dose where it is largest....

late-spill-scout. Late spill on the recipe rose as the anti weight annealed, and holding the anti weight high lowered it at a cost to removal.



Along training, by condition. *One column per condition, one line per run, at the checkpoints kept at epochs 16, 32 and 64 and then every 32 epochs, and at the end.* ***Top:*** *removal under the full edit.* ***Middle:*** *spill under the full edit; the dashed line is the criterion, with the side over it hatched. Both are net of the median over the other 63 axes of the run, and spill here is at the full dose only.* ***Bottom:*** *the anti weight by slice, from the embedding (darkest) to slice 4 (lightest).*

Up to epoch 64 the three annealed conditions have about the same weight, so any difference in how far each run has got by then is run-to-run variation. A few runs of the recipe and of `flat 0.115` are still climbing, and none of `flat 0.2` is.

After epoch 64 the conditions part. Under the recipe and at 0.115 and 0.14, every run climbs to near 0.9 or holds there. At 0.17 most runs do too, and one (seed 720) peaks at 0.81 and ends at 0.71. At 0.2, a few runs slide back, or sit near 0.5 from early on, while the weight on every slice stays high (2 and 1 runs end more than 0.1 under their best, against 0 on the recipe).

So a high weight seems to work against holding the op on e_1, on the seeds where the pull holds it less firmly. These runs can't say whether the weight on the late slices matters or the total: the conditions at 0.2 are higher on every slice past the embedding, and mean removal° also orders by the total weight over training.

Removal. The drop on `difference` under the full edit at every position and slice, as a share of the way from the clean score to the target null....

Spill along training is rarest at 0.2: no run of `flat 0.2` crosses the criterion, and one run of `level 0.2` does, at one checkpoint. `level 0.115` crosses it on the most runs (4), once up to 0.12 in the first half of training, and that run comes back down by the end. `flat 0.115` spikes as high once, at epoch 96 while its weight is still near the start on every slice, and that run ends over.

Of the second round, `level 0.17` and `from 0.17` cross on 1 run each, `level 0.17` up to 0.04 and `from 0.17` only to 0.024, and `level 0.14` on 2, one of them the run that ends over, which sits about at the criterion for most of training and creeps up at the end. None of their runs slides by the 0.1 rule, though seed 720 of `level 0.17` comes within a hair of it (0.097 under its best, which it reached at epoch 128), the pattern of the runs at 0.2 in a milder form.

The recipe crosses on 3 runs, one of them for much of the middle of training, and all are back under by the end. So spill at the end understates how often spill gets through along the way, for the recipe as much as for `level 0.115`. The counts do not order by the weight: the recipe crosses on more runs than `flat 0.115` and any weight from 0.14 up, and about as many as `level 0.115`.

Where the anchor sits (E4)

The falloff was chosen by trial, high at the embedding and low at slice 4. Does its shape follow where the op and the other ops sit on e_1?

slice	falloff hold	α, `difference`	range	α, other ops	range
embedding	0.200	0.03	−0.01 to 0.04	0.02	−0.01 to 0.03
slice 1	0.158	0.33	0.26 to 0.36	0.16	0.13 to 0.20
slice 2	0.115	0.55	0.49 to 0.56	0.09	0.07 to 0.11
slice 3	0.073	0.63	0.60 to 0.63	0.07	0.06 to 0.07
slice 4	0.030	0.49	0.43 to 0.49	0.05	0.04 to 0.05

Where the op and the other ops sit on e_1, by slice, at epoch 400. *α at the example answers, as a mean over the eight runs of the recipe, and the range of that mean over the eight conditions. '`difference`" is over the probe contexts of the anchored op, and "other ops" over those of the other six. Beside them, the hold of the falloff at each slice.*

The op sits mostly at slices 2 to 4, peaking at slice 3, and hardly at all on the embedding. The other ops lean on e_1 most at slice 1, at about half of what the op has there, and less with depth. Every condition has the same profile to within a few hundredths, so the falloff did not make it.

The falloff is high where the other ops lean and low where the op sits: it pushes other ops off e_1 hardest at the embedding and slice 1, and least at slices 3 and 4, where the op sits. A weight the same at every slice has to do both jobs with one number, which may be why it works only in a narrow band. These are α° values at the end of training, read beside a falloff set by trial, so they describe the shape and do not show it is the best one.

Alignment. How far a state points along e_1…

Discussion

A weight of 0.2 on every slice is too much. It keeps spill° out, the color embeddings included, but on some seeds removal° sags, and one run in each condition at 0.2 ends with the anchor lost. That fits what the *split-anti-seeds*° experiment found under the older pull, though at eight runs a side the gap in mean removal is not settled by these runs alone. Whether the cost comes from the weight on the late slices or from the total is not separated here.

Spill. The largest drop on any other op, at the dose where it is largest….

Removal. The drop on `difference` under the full edit at every position and slice, as a share of the way from the clean score to the target null….

split-anti-seeds. Under a step pull on slices 2 to 4, an anti hold that falls off with depth kept the anchor on all eight runs, while holds of 0.2 and 0.1 at every slice each lost one run to the query answer, and the high one removed less.

A weight of 0.115 on every slice, the mean of the falloff, does about as well as the recipe on what we can measure at eight runs. `level 0.115` matches it on removal, and `flat 0.115` is a little lower, within the margin on its mean but not clearly. Each ends with one run over the spill criterion where the recipe has none, and the recipe has its own runs that cross the criterion along training.

Between the two, a constant 0.17 meets the bar we set for doing about as well as the falloff: removal within 0.05, no anchor lost, and no run over the spill criterion at the end. It also has no run that misses the task gate, and it crosses the criterion along training on one run, against 3 on the recipe. That makes it the simplest weight here that meets the bar: one number, the same at every slice and every epoch, with only the end anneal it shares with the anchor. An anneal to 0.115 from 0.17 meets the bar too, which fits the start not needing to be as high as 0.25. Both weights were chosen after the first round was read and scored on the same eight seeds, so neither pass has been checked on fresh ones.

So these runs do not show that the falloff is needed, or the high start. A constant weight can be too high: at 0.2 removal sags on some seeds. Whether it can be too low is less clear. From 0.115 to 0.14 each condition ends with one run in eight over the criterion, which these runs cannot tell from none, and `from 0.17` holds 0.115 with no run over. So one run is all that separates `level 0.17` from `level 0.14` on the bar.

These are eight runs a side, so `level 0.17` meeting the bar is not the same as matching the recipe. Its lowest run (0.71) is lower than any of the recipe, its interval on removal reaches past the margin, and its mean sits between those of 0.14 and 0.2, so it may already have a little of the cost seen at 0.2.

No run that started below 0.25 lost the anchor early (none of forty), and α° settled as fast as with the high start. That bounds the early loss rate only loosely, and every one of those weights is above the anchor weight, a milder case than the one in D2.1. The soft start of the pooling temperature° may be what keeps the pull off the syntax tokens now, though no condition here goes without it.

Alignment. How far a state points along e_1…

Pooling temperature. How sharply the anchor pull picks its positions….

References

split-anti-seeds

Under a step pull on slices 2 to 4, an anti hold that falls off with depth kept the anchor on all eight runs, while holds of 0.2 and 0.1 at every slice each lost one run to the query answer, and the high one removed less.

late-pull

With the pull moved off the embedding, an anti hold of 0.03 at every slice let the color embeddings lean onto e_1 and block 1 spill, and higher holds reduced both at some cost to removal.

embedding-lean

The spill of the older recipe came from a color embedding standing in for the op: the e_1 component of the color embeddings followed their lightness.

loss-rate

With a soft start for the pooling temperature, no run of forty lost the anchor, against one of forty without it.

soft-pool-scout

Keeping the pool over slices 1 to 3 at a sharp temperature removed the early spill, which entered through block 1.

fixed-confirm

`τ_ℓ fixed` kept the anchor on all eight runs at fresh seeds with removal near 0.9 and no run over the spill criterion, and became the recipe of record.

late-spill-scout

Late spill on the recipe rose as the anti weight annealed, and holding the anti weight high lowered it at a cost to removal.

paired-runs

Two runs at one seed under different recipes stay partly alike in the task score through epoch 64, but are only weakly alike in removal and no more alike than chance in spill.

ex-2.1.11

In D2.1, an anti weight held at its low value from the start let the pull latch onto a syntax token, and one held at its high value throughout left the response less graded.

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1. Welch t interval: a confidence interval for a difference of two means that does not assume the two samples have equal variance. ↩